

Acting on the review: provenance, any-node models, and honest pan-India numbers

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Where we are, and what this tackled

Done so far

- A 4-class base map of India at 10 m with a living hierarchy: split, add, rule-split, merge, retrain, all as crisp Earth-Engine tiles.
- A git-backed zoo of cards, save and resume, the lab's IndiaSAT models plugged in, per-fortnight water, STACD provenance.

This round: acting on the review

- Cross-check the STACD record with the other teams and align it.
- Train the lab's model the way they do.
- Let a user attach a model to any node, not only greenery.
- Measure mining and water honestly, pan-India, on ground truth.
- Add a filter for spurious water; improve acacia and mining.

STACD: cross-checking with the other teams

Susmit checked our record against theirs, flagged that some parameters differ, and was unsure which are optional or mandatory. We compared field by field and aligned ours.

- The mandatory STAC fields already matched. The differences were optional metadata we now adopt for uniformity: the version, a collection, real catalog links, a datetime range, keywords, and the run parameters.
- Their pipeline-specific fields, a per-crown table description and drone-model names, do not apply to our raster output, which carries a class legend instead.
- A couple of format details remain open; following up with Saharsh this week, then re-sending.

Training the lab's model the way they do

A correctness check on the farm, plantation, scrubland model we ported.

- We were training it only on ground truth within forty kilometres of the user's box, so a small box saw a weaker model than the one the lab ships.
- The lab trains it across the whole agro-ecological region. We now do the same: drop the box-local restriction, sample the whole region with a balanced cap per class.
- The tree-against-crop model already trained on its full pan-India asset, so it was left alone.
- Why it is re-trained each time and not stored: it is an Earth-Engine model that trains and runs on Google's servers, so there is no file to save; the same fixed data and seed make it reproducible.

Any model on any node, with a suggestion

A core promise is that a user refines any class, not a fixed one.

- The lab's two models were hard-wired to refine greenery. They can now attach to any node the user selects.
- The card still suggests where it normally goes, greenery, but does not decide for the user.
- Worked example: split greenery by a rule, then refine one of those children with the lab's model; only that child changes, the rest of the map stays.
- A guard refuses a clash cleanly, and the machinery underneath already generalised, so this was a small, safe change.

Is the mining pixel-to-polygon good enough?

The question: does vectorising the mining pixels avoid a heavier, learned segmentation model?
Measured pan-India, outside the framework, on real mine polygons.

Mining, on ground-truth polygons	Precision	Recall	F1
Vectorised polygons vs truth (object level)	0.04	0.16	0.07
Pixel classifier, linear	0.45	0.70	0.55
Pixel classifier, forest with tuned cut	0.61	0.58	0.59

- The vectorised polygons over-fragment and rarely line up with a real mine, so they are not a delineator; a learned segmentation model is the route for true objects.
- The pixel classifier is a usable screen: a forest with a lowered decision cut lifts precision from 0.45 to 0.61, the weak spot, at some cost to recall.

Water: small against large

The water model currently runs on Sentinel-1 radar, with Sentinel-2 optical water indices alongside. Measured on the deployed model, split by the size of the water body.

Deployed model, by water-body size	Precision	Recall	F1
Large bodies	1.00	0.97	0.99
Small bodies	0.85	0.67	0.75

- Large water is essentially solved.
- Small water is the real gap: a third of it is missed, a recall problem, not a spurious one, since dry land is wrongly called water only about two percent of the time.

A correction for spurious water, checked on ground truth

The review asked for a threshold that holds water only where it persists, a correction on the water output. We fold it into how annual water is decided and validate it on the lab's Earth-Engine water ground truth.

Persistence threshold	Water precision	Spurious rate	Small-water recall
Any fortnight	0.80	0.15	0.30
At least two fortnights	0.96	0.02	0.11

- Holding a pixel as water only if it was wet in at least two fortnights cuts spurious water from fifteen percent to two, and lifts precision to 0.96.
- The cost lands on small and seasonal water, so a single global threshold is blunt: it argues for the two-classifier design, a lenient first pass that lets seasonal water through, then a stricter one inside water bodies.

Acacia: how many crowns, and a fair filter

- We have 336 acacia crowns and 576 non-acacia. Every crown is a single tree, a median of twenty-seven square metres.
- That is smaller than one ten-metre pixel, so each crown is a mixed pixel, tree blended with the ground around it. This is why the split is hard on this embedding.
- The review's ten-by-ten filter, taken literally, drops ninety-eight percent of the data. Instead we drop only degenerate slivers under fifteen square metres and keep the rest, 296 and 498 crowns.
- So the filter is really a diagnosis: our acacia ground truth is thin and sub-pixel.

Acacia: improving it, and the ceiling

Configuration	Precision	Recall	F1
Linear, single year (baseline)	0.58	0.82	0.68
Linear, several years pooled	0.59	0.84	0.69
Forest, several years pooled	0.68	0.74	0.71

- The proven levers, more years and a non-linear model, lift F1 from 0.68 to 0.71 and accuracy from 0.72 to 0.78, mostly by raising precision. Real, but bounded: the input is still a mixed pixel.
- The true ceiling-raiser is higher-resolution features, Tessera or drone imagery encoded with DINO, which is external. Nothing here is solved; it is honest and improved.

Thank you